

# Calibration Behavior: 2024 vs. 2026

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## Overview

This report compares longitudinal calibration behavior between the 2024 and 2026 calibration datasets.

The analysis reproduces the drift analysis originally applied to the 2024 calibration data. For each anchor, calibration is compared with its calibration on the first day of the observation period.

Two measures of drift are examined:

1. **Instantaneous drift:** the absolute proportional difference between an anchor's current calibration and its calibration on Day 0.
2. **Maximum drift:** the largest instantaneous drift observed up to each point in time.

The analysis also examines the number of anchors that exceed 1%, 2%, 5%, 10%, and 20% deviation thresholds.

For the 2026 data, calendar dates are preserved throughout the analysis. Thus, the `days` variable represents actual elapsed calendar days rather than the number of calibration files retained after missing days were removed.

## 1. Load Data

The 2024 calibration data are read from individual CSV files. Each filename contains the date of retrieval, which is retained as the observation date, alongside the pre-computed calibration value.

The 2026 data requires us to calculate calibration value via given slope and intercept values.

The calibration value is calculated as:

$$y = 100m + b$$

where (m) is the slope and (b) is the intercept.

Table 1: Dataset Summaries

| year | anchors | observations | first_date | last_date  | calendar_days |
|------|---------|--------------|------------|------------|---------------|
| 2024 | 659     | 19111        | 2024-10-21 | 2024-11-18 | 28            |
| 2026 | 444     | 35520        | 2026-01-23 | 2026-04-14 | 81            |

Only anchors with complete observations across all retained calibration dates are used in the final comparison

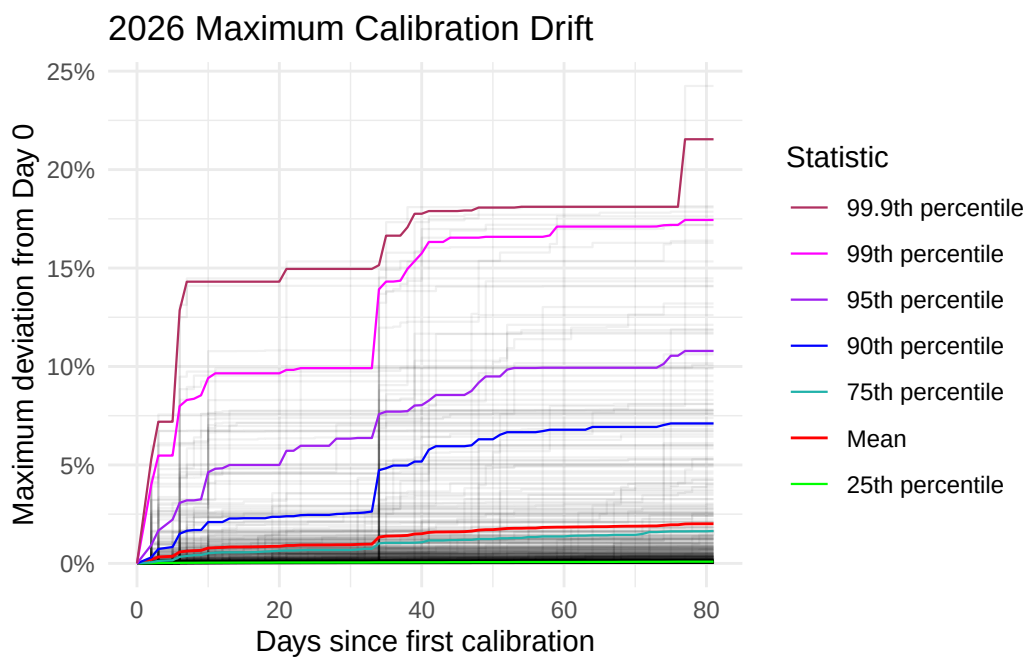
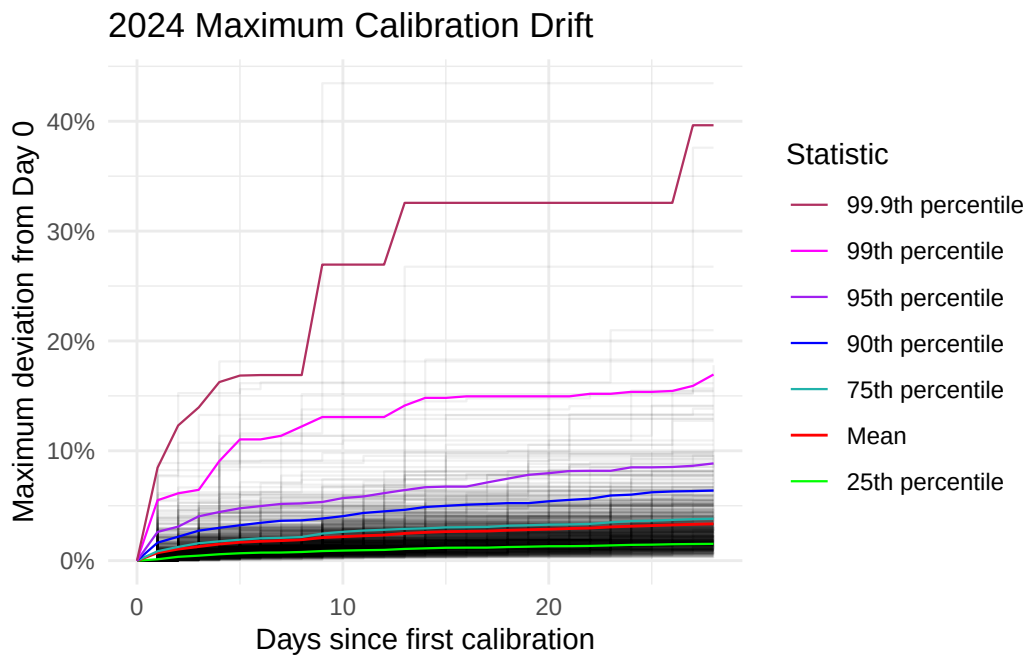
## 2. Calculating Drift

The following calculation is equivalent to Zach’s original 2024 analysis.

For each anchor:

$$drift_t = \frac{|calibration_t - calibration_0|}{calibration_0}$$

### 3. Maximum Drift



### 3.1 90-99.9th Percentile Anchor Summary

- 653 anchors were in the range at least once in 2024 data
- 444 anchors were in the range at least once in 2026 data
- 276 anchors were in the range at least once in both 2024 and 2026 data
- 185 anchors occupied the range only one time each in both 2024 and 2026 data
- 91 anchors occupied the range more than one time each between 2024 and 2026 data

### 3.2 Geographic Context of IDs Associated with repeated ASNs in the 90-99.9 Range

Below, we take a look at the 91 IDs that appeared in the 90-99.9th percentile range more than one time each between the 2024 and 2026 datasets. After joining geographic context to these anchors `cleaned_persistent_90_999_both`, we find that several ASN IDs, cities, and countries are associated more than once with Anchor IDs consistently in the 90-99.9th percentile range. Below are tables for which there was more than anchor association.

Table 2: ASN Frequency

| asn    | anchors |
|--------|---------|
| 14061  | 4       |
| 16276  | 4       |
| 3491   | 3       |
| 57695  | 3       |
| 36236  | 2       |
| 197540 | 2       |

AS14061 belongs to DigitalOcean, LLC, registered in the United States, and hosts 49.9% of its IPv4 share in the United States, 12.4% in Germany, 10.8% in the Netherlands, and 10.3% in Singapore. The rest of the percentages reported are below 10% and are reported to be in the United Kingdom, India, Canada, and Australia. The same countries are listed with different percentages for IPv6. [Link for ASN 14061](#)

AS16276 belongs to OVH SAS, registered in France. It hosts 55.5% of its IPv4 share in France, 17.5% in Canada, and less than 10% in the United States, Germany, the United Kingdom, Poland, Singapore, and Australia. It hosts less than 1% in India, Italy Ireland, Netherlands, Belgium, and Hong Kong. The same countries are listed for IPv6, with different percentages. [Link for ASN 16276](#)

AS3491 belongs to PCCW Global (HK) Ltd., registered in Hong Kong. It hosts 32% of its IPv6 share in Hong Kong, 28.9% in South Africa, 28.6% in Mozambique, and less than 3% in the United States, Singapore, Japan, and <1% in Germany, France, Taiwan, South Korea,

United Kingdom, Australia, Netherlands, Mexico, Sweden, and Kenya. The same countries occupy the shares of IPv4 addresses, with different percentages. [Link for ASN 3491](#)

AS57695 belongs to Misaka Network, Inc., and is registered in the the United States. It is registered in the United States, but does not maintain any IP addresses in the country. It hosts its IPv4 equally amongst the Netherlands, France, Singapore, Brazil, Switzerland, and Germany at 16.7%. For IPv6 addresses, the same countries are listed, but 48.3% of the share is associated with the Netherlands, 48.2% with France, 3.1% with Singapore, and the remaining countries are associated with less than 1%. [Link for ASN 57695](#)

AS36236 belongs to NetActuate, Inc, registered with the United States. 68.6% of its IP share is in the United States, and less than 5% are in 20 different countries. The IPv6 share is similar, with the United States at 65.8% South Africa, Ghana, and Kenya above 9%, and the same IPv4 listed countries for the remaining share. [Link for ASN 36236](#)

AS197540 belongs to netcup GmbH, registered in Germany. Germany dominantes its IPv4 share with 89.2%, Austria represents 9.8%, and the Netherlands the remaining 1%. The same three countries constitute the IPv6 share. [Link for ASN 197540](#)

Table 3: Country Frequency

| country | anchors |
|---------|---------|
| DE      | 16      |
| US      | 14      |
| SG      | 8       |
| AT      | 7       |
| FR      | 6       |
| IT      | 6       |
| CZ      | 5       |
| GB      | 5       |
| NL      | 3       |

This table lists Germany, the United States, Singapore, Austria, France, Italy, Czechia, United Kingdom, and the Netherlands as being associated with more than 2 anchors that occupy the 90-99.9th percentile consistently. Given the IPv4 and IPv6 shares of the previously identified host network providers, this distribution of countries seems reasonable. The city and continent distributions are available in the code as well, but are omitted.

### 3.3 Flaw with Maximum Calibration Drift Visualization: Day 0 as Ground Truth

If we look at the actual day in range proportions of the anchors included in the figures above, we see that some of them occupied the range 1 time in 2024 and 79 out of 80 times in 2026. What this likely demonstrates is that the day 0 calibration measurement for 2026 was an outlier, and for the 79 out of 80 days, the calibration value returned to its ground truth calibration range. However, what gets recorded in the chart is an occupying of the range strictly with respect to the day 0 measurement.

Take a look at the individual anchors associated with the ASN Frequency Table 2:

Table 4: AS14061 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6968 | 9           | 29       | 1           | 80      | Asia      | IN      | Bangalore |
| 6967 | 1           | 29       | 15          | 80      | Europe    | DE      | Frankfurt |
| 6970 | 1           | 29       | 30          | 80      | Asia      | SG      | Singapore |
| 6971 | 1           | 29       | 3           | 80      | Europe    | GB      | London    |

Table 5: AS16276 Anchors

| id   | in_range_24 | to-<br>tal_24 | in_range_26 | total26 | continent | coun-<br>try | city                 |
|------|-------------|---------------|-------------|---------|-----------|--------------|----------------------|
| 6727 | 4           | 29            | 37          | 80      | Oceania   | AU           | Sydney               |
| 6800 | 3           | 29            | 1           | 80      | Europe    | PL           | Ozarów<br>Mazowiecki |
| 6696 | 1           | 29            | 35          | 80      | Europe    | FR           | Gravelines           |
| 6726 | 1           | 29            | 4           | 80      | Asia      | SG           | Singapore            |

Table 6: AS3491 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 7198 | 27          | 29       | 20          | 80      | Asia      | AE      | Dubai     |
| 7078 | 26          | 29       | 76          | 80      | Oceania   | AU      | Sydney    |
| 7010 | 1           | 29       | 32          | 80      | Asia      | SG      | Singapore |

Table 7: AS57695 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent     | country | city     |
|------|-------------|----------|-------------|---------|---------------|---------|----------|
| 6927 | 15          | 29       | 4           | 80      | North America | US      | Secaucus |
| 6928 | 8           | 29       | 1           | 80      | North America | US      | Reston   |
| 7011 | 1           | 29       | 79          | 80      | North America | US      | Seattle  |

Table 8: AS36236 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent     | country | city    |
|------|-------------|----------|-------------|---------|---------------|---------|---------|
| 6379 | 5           | 29       | 1           | 80      | North America | US      | Durham  |
| 6408 | 1           | 29       | 45          | 80      | North America | US      | Phoenix |

Table 9: AS197540 Anchors

| id   | in_range_24 | total_24 | in_range_26 | total26 | continent | country | city      |
|------|-------------|----------|-------------|---------|-----------|---------|-----------|
| 6581 | 1           | 29       | 27          | 80      | Europe    | DE      | Nuremberg |
| 6799 | 1           | 29       | 32          | 80      | Europe    | DE      | Nürnberg  |

### 3.4 Geographic Context of Single Anchor Associated ASNs in the 90-99.9 Range

look at anchors\_90\_999\_2024 and anchors\_90\_999\_2026 and see if the biggest contributor to the 90-999 range is made of single ASN associated IDs. Map the average percentiles of the repeated ASN IDs. they should be on the lower side. look up the ASNs that are the worst, so single IDs that have the most counts for being in the 90-999 range.

get ASN summary for asns that are associated with 1 anchor only, find the ones with the most counts for being in the range, discuss asns and geography, confirm with country frequency table?

Table 10: ASN Frequency

| asn  | anchors |
|------|---------|
| 719  | 1       |
| 2108 | 1       |
| 2857 | 1       |

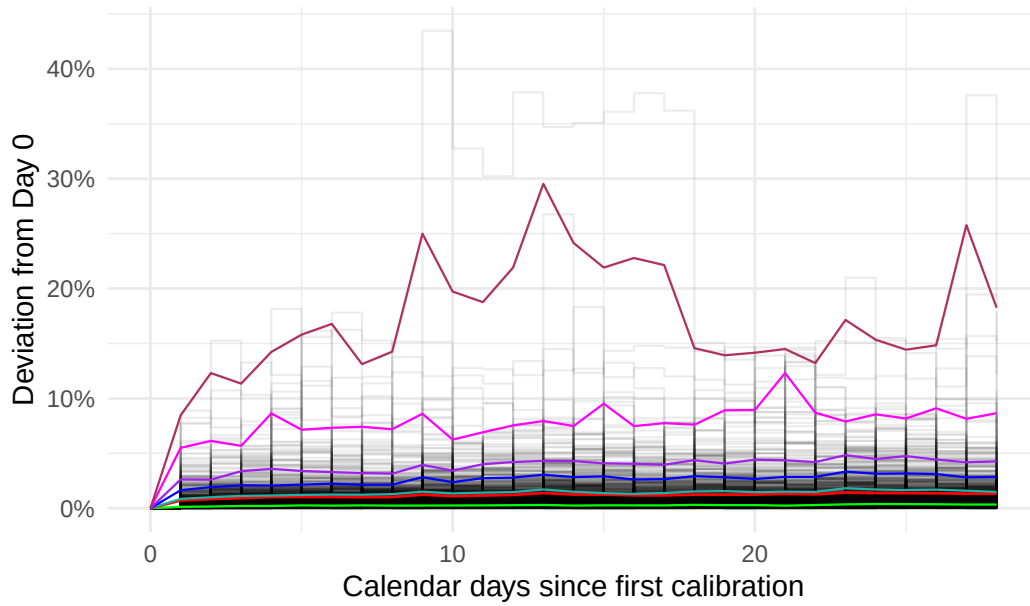
| asn   | anchors |
|-------|---------|
| 6881  | 1       |
| 7752  | 1       |
| 8473  | 1       |
| 12008 | 1       |
| 13101 | 1       |
| 13303 | 1       |
| 14907 | 1       |
| 15699 | 1       |
| 15802 | 1       |
| 16347 | 1       |
| 20473 | 1       |
| 21034 | 1       |
| 21320 | 1       |
| 23428 | 1       |
| 24482 | 1       |
| 24840 | 1       |
| 24940 | 1       |
| 24971 | 1       |
| 25192 | 1       |
| 25248 | 1       |
| 29169 | 1       |
| 29287 | 1       |
| 29423 | 1       |
| 30971 | 1       |
| 31034 | 1       |
| 31510 | 1       |
| 32167 | 1       |
| 33920 | 1       |
| 35492 | 1       |
| 35574 | 1       |
| 36444 | 1       |
| 38001 | 1       |
| 38064 | 1       |
| 39063 | 1       |
| 39617 | 1       |
| 39636 | 1       |
| 39642 | 1       |
| 39835 | 1       |
| 39878 | 1       |
| 41000 | 1       |
| 41095 | 1       |



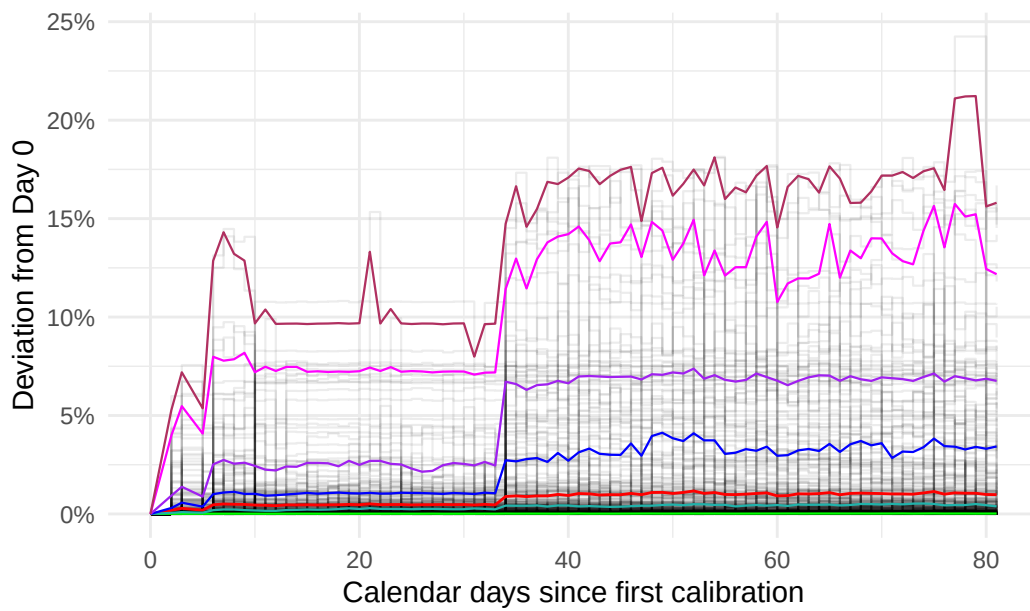
| asn    | anchors |
|--------|---------|
| 41327  | 1       |
| 46997  | 1       |
| 47295  | 1       |
| 48362  | 1       |
| 48943  | 1       |
| 50030  | 1       |
| 53698  | 1       |
| 54316  | 1       |
| 56381  | 1       |
| 59678  | 1       |
| 59715  | 1       |
| 60396  | 1       |
| 60841  | 1       |
| 62310  | 1       |
| 62416  | 1       |
| 62513  | 1       |
| 199188 | 1       |
| 199399 | 1       |
| 199610 | 1       |
| 200187 | 1       |
| 201333 | 1       |
| 202196 | 1       |
| 202297 | 1       |
| 203329 | 1       |
| 204092 | 1       |
| 206186 | 1       |
| 209097 | 1       |
| 216444 | 1       |
| 395823 | 1       |

## 4. Instantaneous Drift

2024 Instantaneous Calibration Drift



2026 Instantaneous Calibration Drift



## 4.1 Percentile Distribution

Following Zach's lead, I applied the Instantaneous Calibration Drift visualization on the 2024 and 2026 data. In the 2024 figure, anchor calibration drift is concentrated towards 0% and mainly seems to decrease in density between 2.5% to 7.5% deviation. The remaining fraying seems to happen between 7.5% and 20% deviation primarily, and a few outlier anchors extend above 20%, 30%, and 40%.

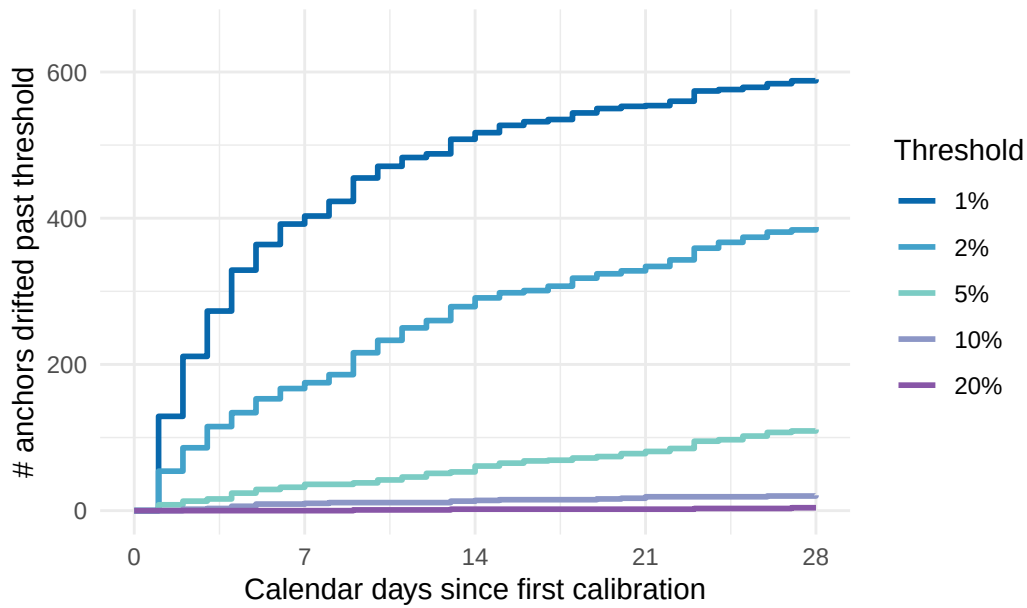
As for the 2026 data, anchor calibration drift seems to be mainly concentrated between 0% and 1%, and spreads similarly to the 2024 2.5% to 7.5% region in the 1% to 4% area between day 0 and day 35. This range seems to extend well into 7.5% after day 35. While the outliers drifted between 5% and 15% deviation before day 35, the outlier group is joined by more anchors and extends past 17.5% consistently between day 35 and the end of the recorded period. While the y-axis deviation scale is inconsistent between the two figures, it is clear that the anchor deviation behavior noticeably shifts in the 2026 data.

Due to the significantly lower number of anchors in the 2026 data and the differing time periods, interpreting these figures alone is not enough. If there were 215 or so more anchors in the 2026 data to correspond with the 659 anchors in the 2024 data, perhaps this shift behavior would have been camouflaged by a complement shift down around day 35. It is not clear that the 2024 figure is completely absent of these kinds of shifts either.

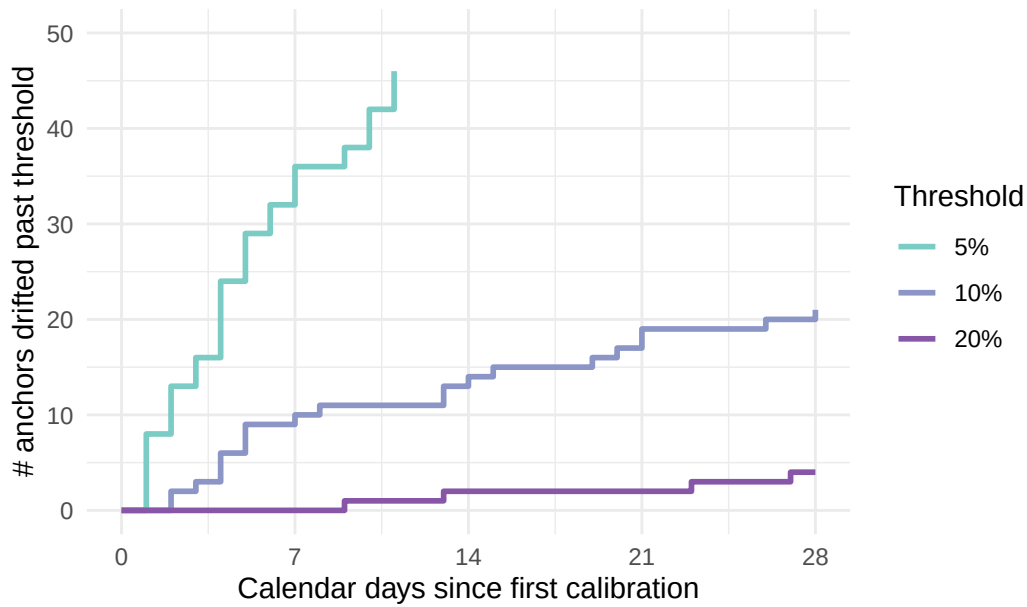
see if the same anchors come up for the 90-999 range

## 5. Maximum Drift Threshold

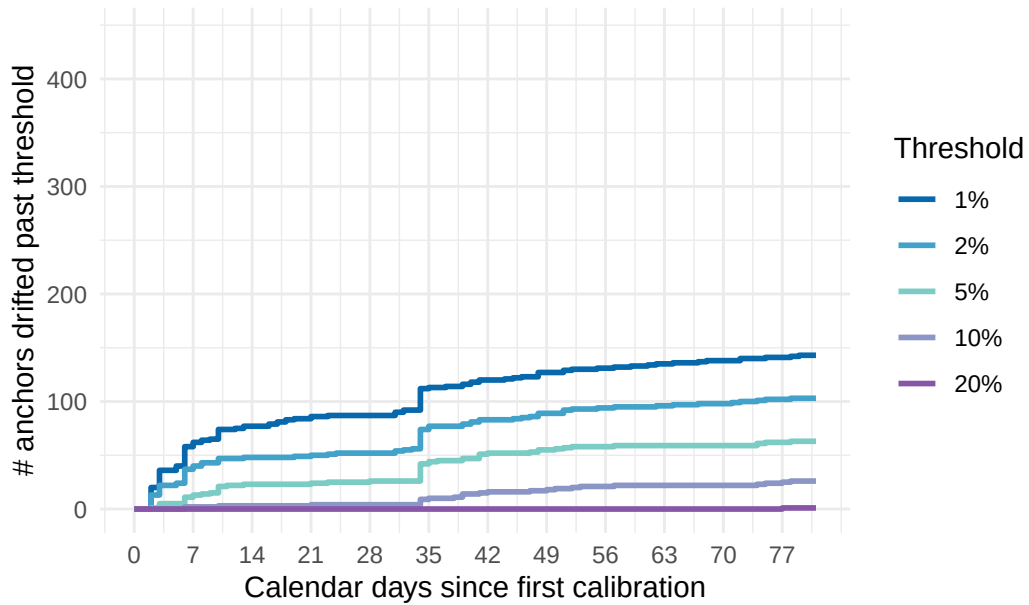
2024: Anchors Ever Exceeding Drift Threshold



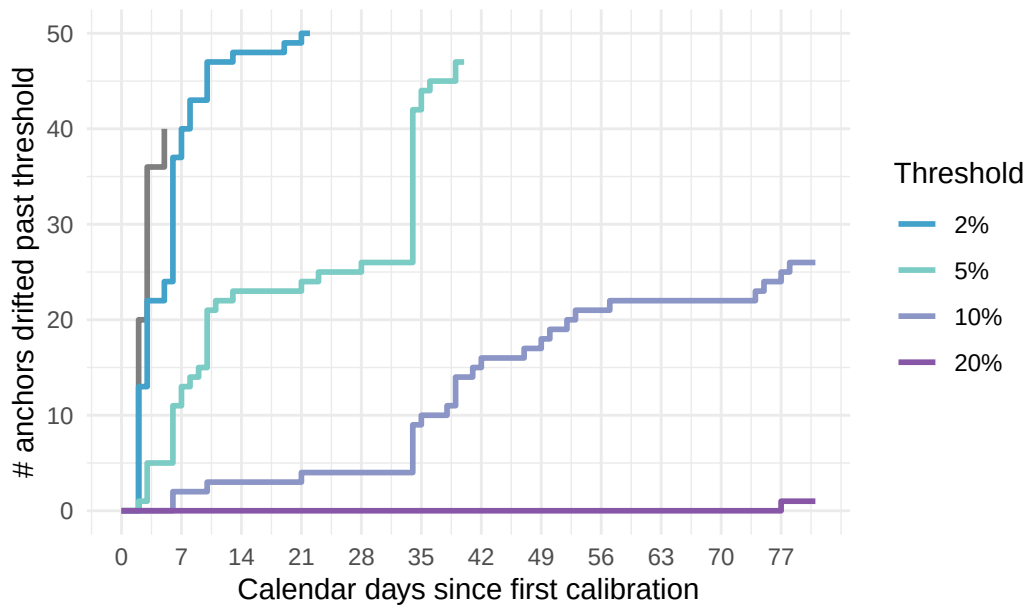
2024: Anchors Ever Exceeding Drift Threshold



2026: Anchors Ever Exceeding Drift Threshold



2026: Anchors Ever Exceeding Drift Threshold



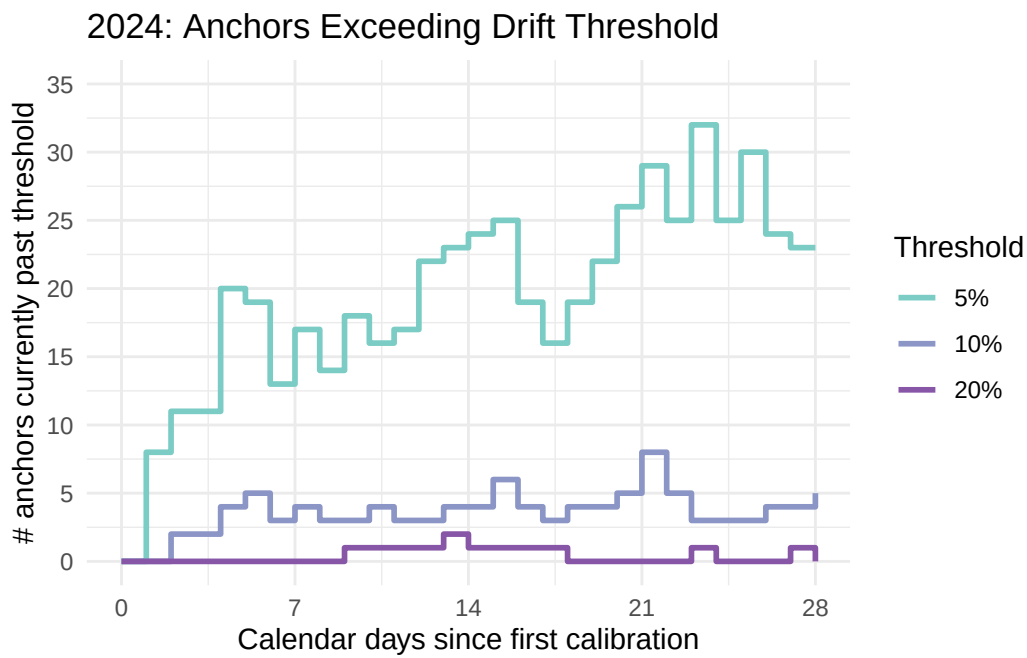
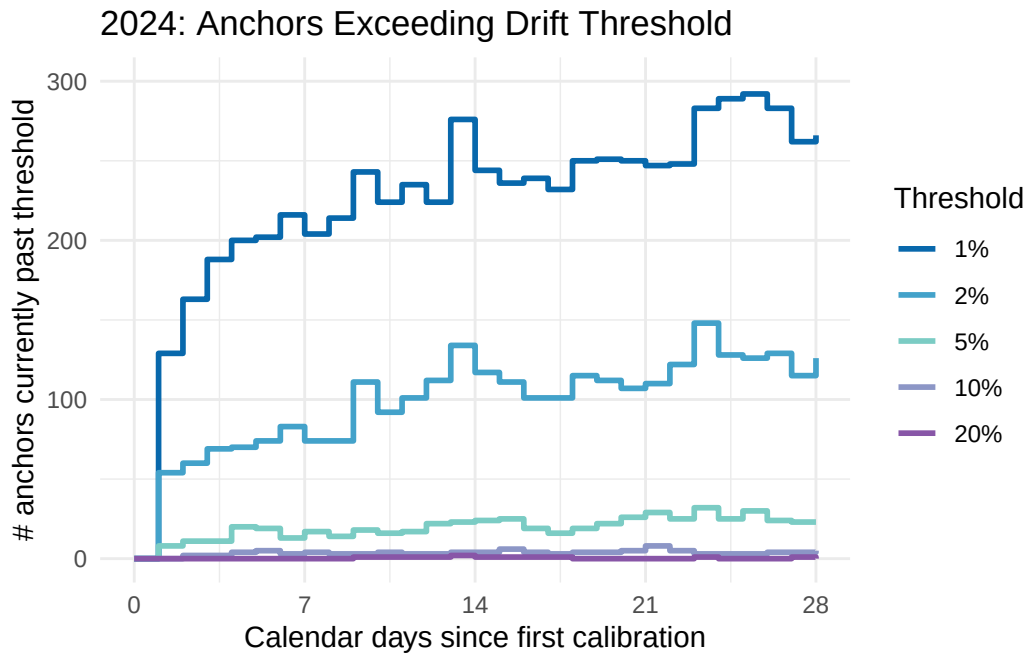
The Maximum Drift Threshold uses percentage thresholds to illustrate the rate at which the anchors exceed 1%, 2% 5%, 10%, and 20% drift. By day 7, nearly 400 of 659 anchors in the 2024 data have exceeded 1% drift, and around 200 of 659 anchors exceed 2% by day 9. In order to improve interpretability, I have adjusted the y axis for examining the 10% and 20% specific thresholds trends. In this figure, we see that about 20 of 659 anchors exceeded 10% drift, and about 5 of 549 exceeded 20% by the end of the recorded period. This visualization is especially convenient for observing the rate at which the anchors exceed each threshold.

For the 2026 data, we use two figures to visualize our threshold details carefully once more. The drift trends for the 2026 data appear to be more conservative than the 2024 data, which is due to the comparatively wider date range and therefore less reliable calibration retrieval (e.g. 9 444 of 899 anchors remain after removing anchors with incomplete series). Nonetheless, the 2026 data demonstrates this shift that we have been noticing around day 35 persists, this time quantifiable as around 10 new anchors reaching 2%, and 5 reaching 5% drift thresholds.

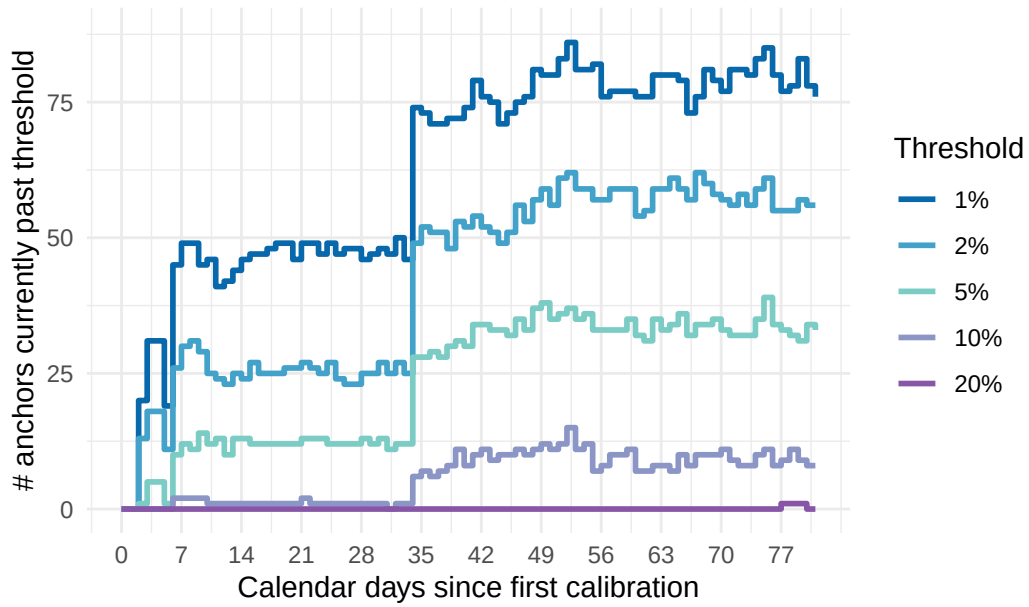
It is especially encouraging to observe that in the 2024 data, the number exceeding the 20% threshold does not exceed 5 anchors, and in 2026, only 1 anchor exceeds 20% after day 77.

At any point, make table for percent drift table for IDs, top will be the highest drift, then the lower drifts, and then lower like ID 8 exceeded 20% 6 and 8 exceeded 10% 6 ad 9 and 0 and 3 exceeded 5% and within each table its ordered by % exceeded

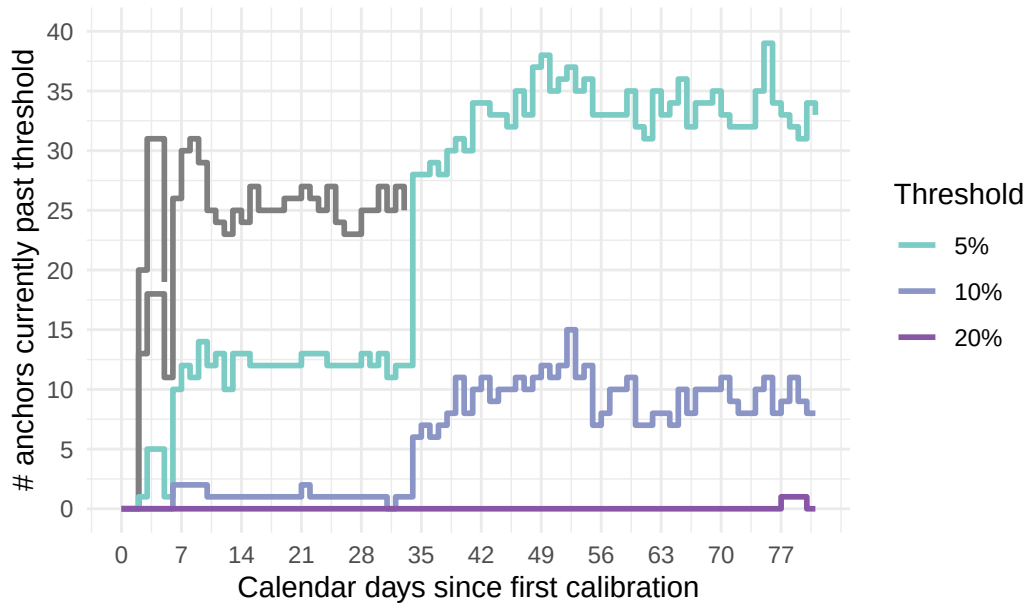
## 6. Instantaneous Drift Threshold



2026: Anchors Exceeding Drift Threshold



2026: Anchors Exceeding Drift Threshold

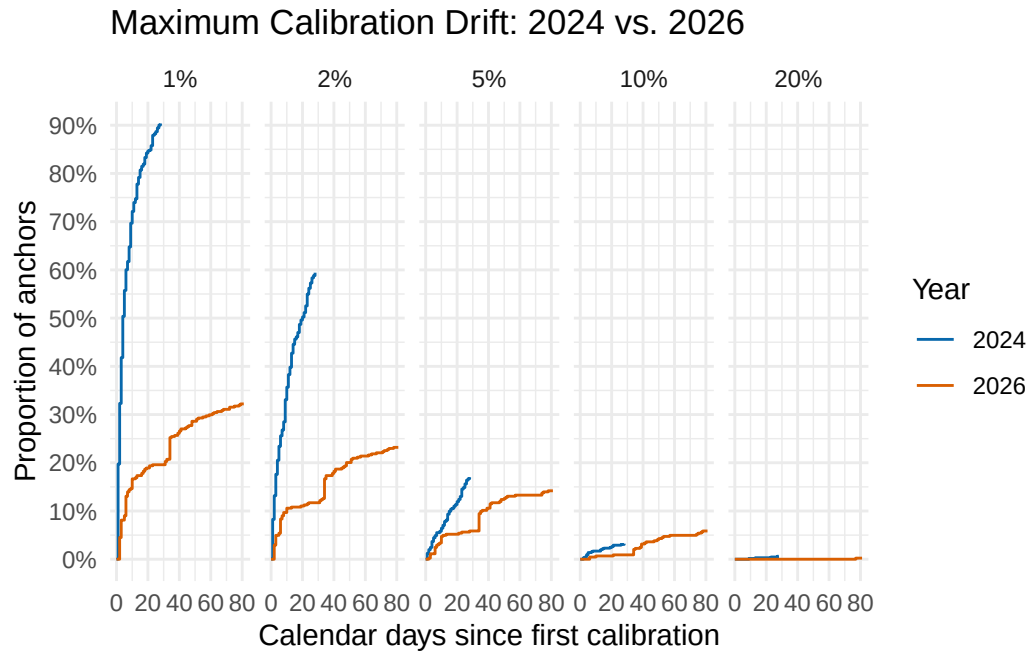




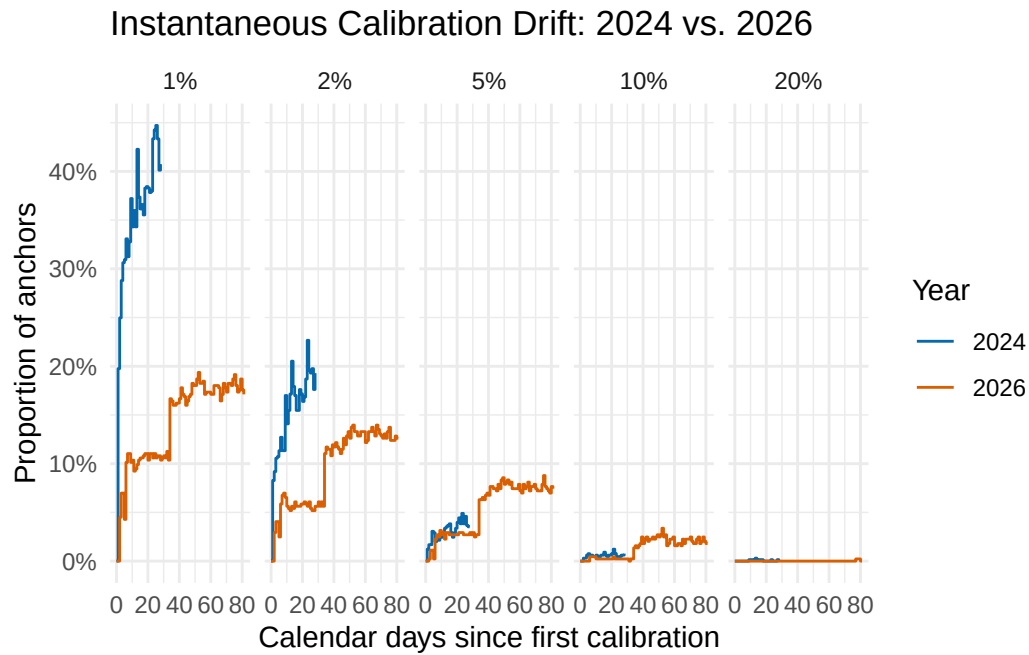
add mean or median values for each threshold line, note value

## 7. Direct 2024 vs. 2026 Comparison

For direct comparison, it is useful to put both years onto the same plot. Because the datasets contain different numbers of anchors, percentages are more informative than raw counts.



Look at the curves and see if there's a relationship between the two



Compare proportion of instantaneous drift , calculate relationship between curves

## 8. Notes for Interpretation

The maximum-drift plots measure whether an anchor has **ever** departed from its initial calibration by a given amount. Consequently, these curves can only increase or remain constant.

The instantaneous-drift plots measure how many anchors are **currently** beyond each threshold. These curves can increase or decrease as anchors move back toward their initial calibration.

A pattern in which maximum drift increases over time while instantaneous drift remains approximately stable suggests that individual anchors experience deviations but that the overall population does not undergo a systematic accumulation of drift.

The 2024 and 2026 comparisons should be interpreted using calendar time. Missing calibration dates in the 2026 dataset remain gaps in the time axis rather than being compressed into consecutive observation numbers.